

Lucy M. Thomas

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Postdoctoral Researcher in Gravitational Wave Astronomy (she/her)

My research focuses on gravitational waveform modelling and Bayesian inference for compact binary coalescences, with an emphasis on developing machine learning-based approaches to improve waveform accuracy and computational efficiency. These advances enable the extraction of maximal astrophysical information from gravitational-wave observations. I am an active member of the LIGO-Virgo-KAGRA (LVK) and Simulating eXtreme Spacetimes (SXS) collaborations.

PROFESSIONAL EXPERIENCE

California Institute of Technology, USA

Postdoctoral Research Scholar Associate

Oct 2023–Aug 2026

- Working within the LIGO Laboratory and Theoretical Astrophysics including Relativity (TAPIR) Research Group

EDUCATION

University of Birmingham, Institute for Gravitational Wave Astronomy, UK

Ph.D. Physics and Astronomy, supervised by Dr. Patricia Schmidt

Sep 2019–Aug 2023

- Thesis title: “*Gravitational Waveform Modelling for Inference of Spin Precession in Binary Black Hole Mergers*”
- Funded by the University of Birmingham and Science and Technologies Funding Council, UK
- PhD Award Date (graduation): 06/12/2023

King’s College London, UK

MSc. Physics, thesis supervised by Dr. Eugene Lim

Sep 2018–Aug 2019

- Thesis title: “*Modelling Critical Phenomena in Scalar Field Collapse*”

Grad. Dip. Physics, thesis supervised by Prof. Mairi Sakellariadou

Sep 2017–Aug 2018

- Thesis title: “*Stochastic Background of Gravitational Waves from Early Universe Phase Transition Bubble Collisions*”

University of Cambridge, UK

M.A. Hons (Cantab) Natural Sciences

Oct 2013–Jul 2017

- Part I Mathematics, Part II Astrophysics
- Project title: “*Chaos in the Encounter Map of the Three-Body Problem*”

POSITIONS OF SCIENTIFIC LEADERSHIP AND SERVICE

- **Co-chair of LVK Waveforms working group** Feb 2025–Current
 - Strategic leadership for waveform development, review and adoption in the collaboration
 - Oversee internal reviews of waveform models before they are integrated into analysis pipelines
 - Represent the Waveforms group interests to the collaboration’s executive bodies
- **Chair of SXS Surrogates working group** Jan 2025–Current
 - Organising and chairing regular meetings
- **Caltech LIGO Seminar Series Lead Organiser** Sep 2023–Current
- **LVK Parameter Estimation Expert**, Senior expert member for event rota and review Sep 2019–Current
- **Postgraduate Student Representative**, Astrophysics and Space Research Group Oct 2019–Aug 2023

INVITED CONFERENCE TALKS

- **University of Illinois, Urbana-Champaign**, Invited Talk 1st Jul 2025
“2G+XG Science”
As part of the Physics at the Xtreme (PAX)-X and Cosmic Explorer Symposium.
- **University of Illinois, Urbana-Champaign**, Invited Panel Discussion 30th Jun 2025
“Waveforms: Status, Challenges and Opportunities”
As part of the Physics at the Xtreme (PAX)-X and Cosmic Explorer Symposium.
- **Institute for Computational and Experimental Research in Mathematics, Brown University**, Invited Talk 3rd Jun 2025
“Public Code Packages to Visualise and Optimise Gravitational Wave Surrogate Models”
As part of the workshop ‘Scientific Machine Learning for Gravitational Wave Astronomy’.
- **APS Global Physics Summit, Anaheim**, Invited Talk 18th Mar 2025
“Building Binary Black Hole Waveform Models Using Numerical Relativity”
As part of the special session ‘20 Years Since of Numerical Relativity’.
- **Royal Astronomical Society, London**, Invited Talk and Panel Discussion 10th Jan 2025
“Machine Learning in Parameter Estimation”
As part of the special discussion workshop ‘Gravitational Wave Data Analysis in the Era of Machine Learning’.
- **Institute for Pure and Applied Mathematics, UCLA**, Invited Talk 14th Nov 2021
emph “A New Effective Precession Spin for the Strong-Field Regime”
Given as invited core participant of the long program “Mathematical and Computational Challenges in the Era of Gravitational Wave Astronomy”.

CONTRIBUTED CONFERENCE TALKS

- **APS Global Physics Summit, Anaheim**, Contributed Talk 18th Mar 2025
“Optimizing Neural Network Surrogate Models: Application to Black Hole Merger Remnants”
- **APS April Meeting, Sacramento**, Contributed Talk 5th Apr 2024
“Constraining Precession in the Black Hole Binary Population with Next-Generation Ground-Based Gravitational Wave Detectors”
- **BritGrav22**, Contributed Talk (virtual) 4th Apr 2022
“A Precessing, Higher-Mode Surrogate Waveform Model Using Neural Networks”
Winner of Best Student Talk
- **APS April Meeting**, Contributed Talk (virtual) 18th Apr 2021
“Modelling Precessing Higher Modes in the Strong-Field Regime”
- **BritGrav21**, Contributed Talk (virtual) 12th Apr 2021
“Modelling Precessing Higher Modes in the Strong-Field Regime”

INVITED SEMINARS

- **University of Southampton, Southampton** 9th Jan 2025
“Machine Learning Applications to Gravitational Wave Modelling”
Southampton Theory Astrophysics and Gravity (STAG) Seminar Series
- **University of Birmingham, Birmingham** 14th Dec 2023
‘Inferring the Spins of the Black Hole Binary Population with Next-Generation Ground-Based Gravitational Wave Detectors’
Astrophysics Seminar Series
- **University of Glasgow, Glasgow** 1st Dec 20223
“SEOBNR_v4PHM_4dq2: Faster Precessing, Higher Multipole Waveforms with Neural Networks”
Given at the Institute for Gravitational Research (IGR) Data Analysis Group Meeting
- **California Institute of Technology, Pasadena** 10th Nov 2021
“A New Effective Precession Spin for the Strong-Field Regime”
Given at the Caltech LIGO Laboratory Data Analysis Group Meeting

TEACHING AND MENTORING EXPERIENCE

- **Undergraduate Research Internship Mentor (START)**, Caltech 25th Aug 2025–Current
Mentoring community college undergraduate student Katina Carranza for a year-long part-time internship project as part of a national NSF program to match community college students with research mentors.
- **Undergraduate Summer Student Mentor (SURF)**, Caltech 17th Jun 2025–22nd Aug 2025
Mentored one undergraduate student Sam Johar for a 10-week project assessing assumptions for eccentric and precessing waveform modelling.
- **Undergraduate Summer Student Mentor (SURF)**, Caltech 18th Jun 2024–23rd Aug 2024
Mentored an undergraduate student Isaiah Taylor 10-week summer project on eccentric and precessing inspiral waveform model generation. Co-mentored 2 other undergraduate summer students (Sena Kalabalık and Andres Nava) on projects on symbolic regression for trends in gravitational wave data, and statistical methods for glitch impact analysis.
- **Undergraduate Summer Student Lecturer**, Caltech 18th Jun 2024–23rd Aug 2024
Gave two interactive lectures for undergraduate summer students entitled “Using Git for Research” and “Waveform Modelling for Compact Binary Coalescences”.
- **Graduate Student Demonstrator**, University of Birmingham 1st Sep 2020–30th Jul 2023
Helped teach and assess numerous undergraduate courses in Python computing, mathematics for first year undergraduate physicists, and physics labs.

AWARDS, PRIZES AND FUNDING

- **Grant Proposal Contributor** NSF Award Number 2207758
“Gravitational Wave Physics and Astrophysics with LIGO” Significant written contribution to the awarded grant (roughly 1/3 of the proposal content)
- **Winner 2022** BritGrav22 Best Student Talk
“A Precessing, Higher-Mode Surrogate Waveform Model Using Neural Networks” - [watch the video](#)
- **Winner 2022** University of Birmingham School of Physics Student Paper of the Year Award
“New Effective Precession Spin for Modeling Multimodal Gravitational Waveforms in the Strong-Field Regime”
- **Overall Winner 2021** University of Birmingham Research Poster Conference Competition
“You Spin Me Right Round: Modelling ‘Wobbly’ Gravitational Waves from Binary Black Holes” - [view the poster](#)
- **Winner 2020** UK 3 Minute Thesis Competition
“Gravity is Pathetic! Shedding Light on Gravity with Models of Spinning Black Hole Mergers” - [watch the video](#)